

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TRANSMISSION CONTROL MODULE (TCM) - HYBRID ELECTRIC VEHICLE

WARNINGS:

- A risk assessment must be performed before any work is undertaken.
- All hybrid electric vehicle work must be performed by a suitably qualified person.
- Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.
- The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.
- If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)
- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Transmission Control Module (TCM) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Diagnostics - SDV6 3.0L Diesel - Hybrid](#)

Electric Vehicle (307-01C Automatic Transmission/Transaxle - Vehicles With: 8HP70 8-Speed - Hybrid Electric Vehicle - Automatic Transmission – AWD, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B108E-02	Display - General signal failure	▪ Gear display signal failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs. Road test the vehicle and check that the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module
C1A88-64	High Pressure Switching Valve #1 - Signal plausibility failure	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0560-21	System Voltage - Signal amplitude < minimum	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and test the transmission control module power and ground circuits for open circuit, high resistance
P0562-1C	System Voltage Low - Circuit voltage out of range	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and test the transmission control module power and ground circuits for open circuit, high resistance

P0562-21	System Voltage Low - Signal amplitude < minimum	▪ Battery/charging system fault ▪ Transmission control module power or ground circuit open circuit, high resistance	▪ Refer to the relevant section of the workshop manual and test the battery and charging system ▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and test the transmission control module power and ground circuits for open circuit, high resistance
P0563-22	System Voltage High - Signal amplitude > maximum	▪ Battery/charging system fault	▪ Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the relevant section of the workshop manual and test the battery and charging system
P0601-41	Internal Control Module Memory Checksum Error - General checksum failure	▪ Software failure ▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system check and install the latest relevant level of software to the transmission control module ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-04	Control Module Processor - System internal failures	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-11	Control Module Processor - Circuit short to ground	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-61	Control Module Processor - Signal	▪ Transmission control module	

	calculation failure	failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-62	Control Module Processor - Signal compare failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0606-64	Control Module Processor - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P060A-61	Internal Control Module Monitoring Processor Performance - Signal calculation failure	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-04	TCM Processor - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-06	TCM Processor - Algorithm based failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

			fault persists, install a new transmission control module
P0613-11	TCM Processor - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-12	TCM Processor - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-13	TCM Processor - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-15	TCM Processor - Circuit short to battery or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0613-18	TCM Processor - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P061B-02	Internal Control Module Torque Calculation Performance - General signal failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P062F-04	Internal Control Module EEPROM Error - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0641-64	Sensor Reference Voltage A Circuit/Open - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0642-21	Sensor Reference Voltage A Circuit Low - Signal amplitude < minimum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0643-22	Sensor Reference Voltage A Circuit High - Signal amplitude > maximum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0657-13	Actuator Supply Voltage A Circuit / Open - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0657-64	Actuator Supply Voltage A Circuit / Open - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0658-11	Actuator Supply Voltage A Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0659-12	Actuator Supply Voltage A Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

			transmission control module
P0666-64	Control Module Internal Temperature Sensor "A" Circuit - Signal plausibility failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0667-01	PCM / ECM / TCM Internal Temperature Sensor A Range/Performance - General electrical failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0668-21	PCM / ECM / TCM Internal Temperature Sensor A Circuit Low - Signal amplitude < minimum	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control

			module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0669-22	PCM / ECM / TCM Internal Temperature Sensor A Circuit High - Signal amplitude > maximum	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AC-01	PCM / ECM / TCM Internal Temperature Sensor B Range/Performance - General electrical failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AC-64	PCM / ECM / TCM Internal Temperature Sensor B Range/Performance - Signal plausibility	Transmission control module failure	

	failure		NOTE: Consider environmental conditions before condemning the transmission control module ▪ Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AD-22	PCM / ECM / TCM Internal Temperature Sensor B Circuit Low - Signal amplitude > maximum	▪ Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module ▪ Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P06AE-21	PCM / ECM / TCM Internal Temperature Sensor B Circuit High - Signal amplitude < minimum	▪ Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module ▪ Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the

			DTCs and retest. If the fault persists, install a new transmission control module
P0700-02	Transmission Control System (MIL Request) - General signal failure	Transmission system fault	Investigate other DTCs first. Clear the DTCs and road test the vehicle. Re-read DTCs
P0700-83	Transmission Control System (MIL Request) - Value of signal protection calculation incorrect	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0702-98	Transmission Control System Electrical - Component or system over temperature	Insufficient transmission cooling	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the automatic transmission cooling system
P0710-04	Transmission Fluid Temperature Sensor A Circuit - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-13	Transmission Fluid Temperature Sensor A Circuit - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-27	Transmission Fluid Temperature Sensor A Circuit - Signal rate of change above threshold	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the

			manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0710-64	Transmission Fluid Temperature Sensor A Circuit - Signal plausibility failure	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0712-11	Transmission Fluid Temperature Sensor A Circuit Low - Circuit short to ground	Transmission control module failure	NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0713-12	Transmission Fluid Temperature Sensor A Circuit High - Circuit short to battery	Transmission control module failure	

			NOTE: Consider environmental conditions before condemning the transmission control module Allow the transmission control module to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0716-61	Turbine/Input Shaft Speed Sensor A Circuit Range/Performance - Signal calculation failure	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. Test the operation of the stop/start system. If the fault persists, install a new transmission control module
P0720-12	Output Shaft Speed Sensor Circuit - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0720-14	Output Shaft Speed Sensor Circuit - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0721-02	Output Shaft Speed Sensor Circuit	Output shaft speed sensor general signal	Check the transmission control module connector and the power/ground circuits to the

	Range/Performance - General signal failure	failure	transmission control module. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Clear the DTC and retest. If the problem persists, renew the transmission control module as required
P0721-27	Output Shaft Speed Sensor Circuit Range/Performance - Signal rate of change above threshold	Signal rate of change above threshold	NOTE: This DTC may be set by an anti-lock braking system fault allowing the brakes to lock during braking Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0721-62	Output Shaft Speed Sensor Circuit Range/Performance - Signal compare failure	- Anti-lock brake system fault - Transfer Case Control Module fault - Transmission control module failure - Output shaft speed sensor signal compare failure	- Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P0721-64	Output Shaft Speed Sensor	Transmission control module	Using the manufacturer approved diagnostic system,

	Circuit Range/Performance - Signal plausibility failure	failure	clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P0729-07	Gear 6 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0730-00	Incorrect Gear Ratio - No sub type information	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0731-07	Gear 1 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0732-07	Gear 2 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0733-07	Gear 3 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the

			manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0734-07	Gear 4 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0735-07	Gear 5 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0736-07	Reverse Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0736-64	Reverse Incorrect Ratio - Signal plausibility failure	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0745-64	Pressure Control Solenoid A - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

			fault persists, install a new transmission control module
P0745-93	Pressure Control Solenoid A - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0745-94	Pressure Control Solenoid A - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-11	Pressure Control Solenoid A Electrical - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-12	Pressure Control Solenoid A Electrical - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0748-13	Pressure Control Solenoid A Electrical - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P076F-07	Gear 7 Ratio Incorrect - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P077E-02	Transmission Fluid Temperature Measurement	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

	System - Multiple Sensor Correlation - General signal failure		fault persists, install a new transmission control module
P0780-62	Shift Malfunction - Signal compare failure	Signal compare failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0780-93	Shift Malfunction - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0780-94	Shift Malfunction - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0791-12	Intermediate Shaft Speed Sensor A Circuit - Circuit short to battery	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0791-14	Intermediate Shaft Speed Sensor A Circuit - Circuit short to ground or open	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0792-27	Intermediate Shaft Speed Sensor A Circuit Range/Performance - Signal rate of change above threshold	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0792-64	Intermediate Shaft Speed Sensor A Circuit Range/Performance - Signal plausibility failure	- Anti-lock brake system fault - Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P0795-04	Pressure Control Solenoid C - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P07D9-07	Gear 8 Incorrect Ratio - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission

P07DC-07	Incorrect Shift from Gear 1 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DD-07	Incorrect Shift from Gear 2 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DE-07	Incorrect Shift from Gear 3 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07DF-07	Incorrect Shift from Gear 4 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E0-07	Incorrect Shift from Gear 5 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new

			automatic transmission
P07E1-07	Incorrect Shift from Gear 6 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E2-07	Incorrect Shift from Gear 7 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P07E3-07	Incorrect Shift from Gear 8 - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P0801-94	Reverse Inhibit Control Circuit - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0810-04	Clutch Position Control Error - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Ensure the displayed gear matches the engaged gear. If the fault persists, install a new transmission control module
P084F-	Park/neutral Switch	Transmission	Using the manufacturer

11	Output Circuit - Circuit short to ground	control module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-12	Park/neutral Switch Output Circuit - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-13	Park/neutral Switch Output Circuit - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-14	Park/neutral Switch Output Circuit - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P084F-29	Park/neutral Switch Output Circuit - Signal invalid	Starter inhibit signal invalid	Check for correct output signal at transmission control module pin 1 to transmission plug (check in all positions). If the signal appears normal, then check wiring and connectors to the module. If no signal present, renew the transmission control module as required
P0850-02	Park/neutral Switch Input Circuit - General signal failure	Park/neutral switch output circuit failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, install a new transmission control module
P0850-29	Park/neutral Switch Input Circuit - Signal invalid	Park/neutral switch output circuit signal invalid	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and

			disengaging the parking lock several times. If fault persists, renew parklock components as necessary. If the fault persists, install a new transmission control module
P0851-01	Park/neutral Switch Input Circuit Low - General electrical failure	Switch input circuit general signal failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, install a new transmission control module
P0852-01	Park/neutral Switch Input Circuit High - General electrical failure	Switch input circuit general signal failure	Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. Test the parklock mechanism by engaging and disengaging the parking lock several times. If fault persists, install a new transmission control module
P0965-64	Pressure Control Solenoid B Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0965-93	Pressure Control Solenoid B Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0965-94	Pressure Control Solenoid B Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0966-11	Pressure Control Solenoid B Control Circuit Low - Circuit	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

	short to ground		fault persists, install a new transmission control module
P0966-12	Pressure Control Solenoid B Control Circuit Low - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0966-14	Pressure Control Solenoid B Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-13	Shift Solenoid A Control Circuit Range/Performance - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-64	Shift Solenoid A Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-93	Shift Solenoid A Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0972-94	Shift Solenoid A Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0973-11	Shift Solenoid A Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0973-18	Shift Solenoid A Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0974-12	Shift Solenoid A Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0974-19	Shift Solenoid A Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-13	Shift Solenoid B Control Circuit Range/Performance - Circuit open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-64	Shift Solenoid B Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-93	Shift Solenoid B Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0975-94	Shift Solenoid B Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0976-11	Shift Solenoid B Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

			transmission control module
P0976-18	Shift Solenoid B Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0977-12	Shift Solenoid B Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0977-19	Shift Solenoid B Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-21	Shift Solenoid C Control Circuit Range/Performance - Signal amplitude < minimum	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-64	Shift Solenoid C Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-93	Shift Solenoid C Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0978-94	Shift Solenoid C Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0979-	Shift Solenoid C	Transmission	Using the manufacturer

11	Control Circuit Low - Circuit short to ground	control module failure	approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0979-14	Shift Solenoid C Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0980-12	Shift Solenoid C Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0980-19	Shift Solenoid C Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0981-64	Shift Solenoid D Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0981-93	Shift Solenoid D Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0981-94	Shift Solenoid D Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0982-11	Shift Solenoid D Control Circuit Low - Circuit short to ground	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P0982-14	Shift Solenoid D Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0982-18	Shift Solenoid D Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0983-12	Shift Solenoid D Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0983-19	Shift Solenoid D Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-64	Shift Solenoid E Control Circuit Range/Performance - Signal plausibility failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-93	Shift Solenoid E Control Circuit Range/Performance - No operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0984-94	Shift Solenoid E Control Circuit Range/Performance - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0985-11	Shift Solenoid E Control Circuit Low - Circuit short to	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

	ground		fault persists, install a new transmission control module
P0985-14	Shift Solenoid E Control Circuit Low - Circuit short to ground or open	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0985-18	Shift Solenoid E Control Circuit Low - Circuit current below threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0986-12	Shift Solenoid E Control Circuit High - Circuit short to battery	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0986-19	Shift Solenoid E Control Circuit High - Circuit current above threshold	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0A90-92	Drive Motor "A" Performance - Performance or incorrect operation	- Motor generator not operating when requested - Transmission control switch malfunction	- Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the transmission control switch for related DTCs and refer to the relevant DTC index
P0B0D-82	Auxiliary Transmission Fluid Pump Motor Control Module - Alive/sequence counter incorrect / not updated	Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance	NOTE: This DTC applies only to hybrid electric vehicles

		▪ Electric automatic transmission fluid pump failure	▪ Refer to the electrical circuit diagrams and check the electric automatic transmission fluid pump power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric automatic transmission fluid pump
P0B0D-93	Auxiliary Transmission Fluid Pump Motor Control Module - No operation	▪ Electric automatic transmission fluid pump failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new electric automatic transmission fluid pump
P0C23-13	Auxiliary Transmission Fluid Pump Control Module Circuit / Open - Circuit open	▪ Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance ▪ Electric automatic transmission fluid pump failure ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the electric automatic transmission fluid pump power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric automatic transmission fluid pump ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the

			fault persists, install a new transmission control module
P0C24-11	Auxiliary Transmission Fluid Pump Control Module Circuit Low - Circuit short to ground	▪ Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance ▪ Electric automatic transmission fluid pump failure ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the electric automatic transmission fluid pump power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric automatic transmission fluid pump ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0C24-14	Auxiliary Transmission Fluid Pump Control Module Circuit Low - Circuit short to ground or open	▪ Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance ▪ Electric automatic transmission fluid pump failure ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the electric automatic transmission fluid pump power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric automatic transmission fluid pump ▪ Using the manufacturer

			approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0C25-12	Auxiliary Transmission Fluid Pump Control Module Circuit High - Circuit short to battery	▪ Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance ▪ Electric automatic transmission fluid pump failure ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the electrical circuit diagrams and check the electric automatic transmission fluid pump power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new electric automatic transmission fluid pump ▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P0C29-93	Auxiliary Transmission Fluid Pump Driver Circuit Performance - No operation	▪ Electric automatic transmission fluid pump failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new electric automatic transmission fluid pump
P1706-94	High Vehicle Speed Observed in Park - Unexpected operation	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new

			transmission control module
P1707-72	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator stuck open	Actuator stuck open	Ensure that emergency park release has not been pulled and is not sticking. Using the manufacturer approved diagnostic system, clear the DTCs and retest, engaging and disengaging the parking lock several times. Check the parking lock mechanism. If the fault persists, install a new transmission control module
P1707-74	Transfer Case Neutral or Park/Neutral Indication Circuit - Actuator slipping	Actuator slipping	Ensure that emergency park release has not been pulled and is not sticking. Using the manufacturer approved diagnostic system, clear the DTCs and retest, engaging and disengaging the parking lock several times. Check the parking lock mechanism. If the fault persists, install a new transmission control module
P1707-77	Transfer Case Neutral or Park/Neutral Indication Circuit - Commanded position not reachable	Commanded position not reachable	- Ensure that emergency park release has not been pulled and is not sticking - Clear the DTC and test drive the vehicle, engaging and disengaging the parking lock several times. If the problem persists, renew the transmission control module as required - Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance (hybrid electric vehicles only)
P1707-94	Transfer Case Neutral or Park/Neutral Indication Circuit - Unexpected operation	Unexpected operation	- Ensure that emergency park release has not been pulled and is not sticking - Clear the DTC and test drive the vehicle, engaging and disengaging the parking lock several times. If the problem

			persists, renew the transmission control module as required ▪ Electric automatic transmission fluid pump power or ground circuit open circuit, high resistance (hybrid electric vehicles only)
P1712-61	Transmission Torque Reduction Request Signal - Signal calculation failure	▪ Engine system fault ▪ Electric power inverter converter system fault ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P1712-64	Transmission Torque Reduction Request Signal - Signal plausibility failure	▪ Engine system fault ▪ Electric power inverter converter system fault ▪ Transmission control module internal failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the electric power

			inverter converter module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P177A-07	Transmission Friction Element A or B - Mechanical failures	▪ Automatic transmission failure	▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177B-07	Transmission Friction Element A or C - Mechanical failures	▪ Automatic transmission failure	▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177C-07	Transmission Friction Element A or D - Mechanical failures	▪ Automatic transmission failure	▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177D-07	Transmission Friction Element A or E - Mechanical failures	▪ Automatic transmission failure	▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission

P177E-07	Transmission Friction Element B or C - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P177F-07	Transmission Friction Element B or D - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178A-07	Transmission Friction Element B or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178B-07	Transmission Friction Element C or D - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P178C-07	Transmission Friction Element C or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new

			automatic transmission
P178D-07	Transmission Friction Element D or E - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P1A05-85	Desired Engine Speed Signal - Signal above allowable range	- Engine system fault - Electric power inverter converter system fault - Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index - Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P2159-02	Vehicle Speed Sensor B Range/Performance - General signal failure	Missing/invalid data from the anti-lock brake system control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer

			approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
P215B-62	Vehicle Speed / Output Shaft Speed Correlation - Signal compare failure	Missing/invalid data from the anti-lock brake system control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the anti-lock brake system control module/transfer case control module for related DTCs and refer to the relevant DTC index
P258F-02	Torque Management Request Output Signal - General signal failure	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2637-84	Torque Management Feedback Signal A - Signal below allowable range	Torque intervention from engine control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
P2700-07	Transmission Friction Element A Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new

			automatic transmission
P2701-07	Transmission Friction Element B Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2702-07	Transmission Friction Element C Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2703-07	Transmission Friction Element D Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2704-07	Transmission Friction Element E Apply Time Range/Performance - Mechanical failures	Automatic transmission failure	Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2711-94	Unexpected Mechanical Gear Disengagement - Unexpected operation	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module

P2713-04	Pressure Control Solenoid D - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2722-04	Pressure Control Solenoid E - System internal failures	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2784-02	Input/Turbine Speed Sensor A/B Correlation - General signal failure	Sensor general signal failure, multiple input speed signal faults	NOTE: Rectify any speed related DTCs first Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
P2787-4B	Clutch Temperature Too High - Over temperature	Clutch over temperature	Allow the transmission to cool, clear the DTC and retest. Check transmission oil level and check for excessive gear changes during normal driving. Check paddle switch for intermittent operation and renew as required
P2787-98	Clutch Temperature Too High - Component or system over temperature	Component or system over temperature	- Allow the transmission to cool, clear the DTC and retest. Check transmission oil level and check for excessive gear changes during normal driving. Check paddle switch for intermittent operation - Check the transmission oil level and quality, refer to the relevant section of the workshop manual. Clear the DTC and test drive the vehicle using all gears in drive and reverse. Clear the DTC and retest. If the problem persists,

			or if any other gear ratio related DTCs are logged, renew the transmission
P2787-9A	Clutch Temperature Too High - Component or system operating conditions	▪ Insufficient transmission cooling ▪ Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the automatic transmission cooling system ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2793-94	Gear Shift Direction Circuit - Unexpected operation	▪ Transmission control module failure	▪ Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2873-94	Clutch "B" Pressure Engagement Too High - Unexpected operation	▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P2879-	Clutch "B"	▪ Automatic	

93	Performance/Stuck Disengaged - No operation	transmission failure	NOTE: This DTC applies only to hybrid electric vehicles Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P287A-92	Clutch "B" Stuck Engaged - Performance or incorrect operation	Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P287E-13	Engine Disconnect Clutch Solenoid Circuit/Open - Circuit open	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P287F-64	Engine Disconnect Clutch Solenoid	Transmission control module	

	Circuit Range/Performance - Signal plausibility failure	failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P287F-93	Engine Disconnect Clutch Solenoid Circuit Range/Performance - No operation	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P287F-94	Engine Disconnect Clutch Solenoid Circuit Range/Performance - Unexpected operation	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2880-11	Engine Disconnect Clutch Solenoid Circuit Low - Circuit short to ground	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system,

			clear the DTCs and retest. If the fault persists, install a new transmission control module
P2881-12	Engine Disconnect Clutch Solenoid Circuit High - Circuit short to battery	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
P2882-92	Engine Disconnect Clutch Excessive Slippage - Performance or incorrect operation	Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P2882-94	Engine Disconnect Clutch Excessive Slippage - Unexpected operation	- Engine system fault - Electric power inverter converter system fault - Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system,

			check the electric power inverter converter module for related DTCs and refer to the relevant DTC index Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2883-4B	Engine Disconnect Clutch Temperature Too High - Over temperature	- Engine system fault - Electric power inverter converter system fault - Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index - Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Check the automatic transmission cooling system. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2883-98	Engine Disconnect Clutch Temperature Too High - Component or system over	Automatic transmission failure	

temperature

NOTE:

This DTC applies only to hybrid electric vehicles

- Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission

P2884-64

Engine Disconnect Clutch Stuck Open - Signal plausibility failure

- Engine system fault
- Electric power inverter converter system fault
- Transmission control module failure

NOTE:

This DTC applies only to hybrid electric vehicles

- Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
- Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index
- Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module

P2884-72

Engine Disconnect Clutch Stuck Open - Actuator stuck open

- Engine system fault
- Electric power inverter

		converter system fault Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index - Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2885-92	Engine Disconnect Clutch Engagement Fault - Performance or incorrect operation	- Engine system fault - Anti-lock brake system fault - Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles - Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index - Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index - Refer to the relevant section of the workshop manual and check the automatic transmission fluid

			level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
P2885-94	Engine Disconnect Clutch Engagement Fault - Unexpected operation	▪ Engine system fault ▪ Electric power inverter converter system fault ▪ Transmission control module failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the electric power inverter converter module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new transmission control module
P2886-92	Engine Disconnect Clutch Stuck Closed - Performance or incorrect operation	▪ Engine system fault ▪ Electric power inverter converter system fault ▪ Automatic transmission failure	NOTE: This DTC applies only to hybrid electric vehicles ▪ Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approved diagnostic system, check the electric power inverter converter module for torque/speed related DTCs and

			refer to the relevant DTC index Refer to the relevant section of the workshop manual and check the automatic transmission fluid level and quality. Using the manufacturer approved diagnostic system, clear the DTCs and road test the vehicle. If the fault persists, install a new automatic transmission
U0001-81	High Speed CAN Communication Bus - Invalid serial data received	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-82	High Speed CAN Communication Bus - Alive/sequence counter incorrect / not updated	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-83	High Speed CAN Communication Bus - Value of signal protection calculation incorrect	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-86	High Speed CAN Communication Bus - Signal invalid	Invalid data received from another control module via the high speed CAN bus (powertrain)	Using the manufacturer approved diagnostic system, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0001-	High Speed CAN	Missing	Using the manufacturer

87	Communication Bus - Missing message	message from another control module via the high speed CAN bus (powertrain)	approved diagnostic system, check the snapshot data to determine the missing message source control module. Check the relevant control module for related DTCs and refer to the relevant DTC index
U0073-00	Control Module Communication Bus "A" Off - No sub type information	High speed CAN bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the high speed CAN bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0100-87	Lost Communication With ECM/PCM "A" - Missing message	Missing message from the engine control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0140-87	Lost Communication With Body Control Module - Missing message	Missing message from the body control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the body control module for related DTCs and refer to the relevant DTC index
U0300-57	Internal Control Module Software Incompatibility -	Car configuration file mismatch	Using the manufacturer approved diagnostic system, check and up-date the car

	Invalid/incomplete software component	with vehicle specification	configuration file as necessary
U0401-82	Invalid Data Received from ECM/PCM A - Alive/sequence counter incorrect / not updated	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0401-83	Invalid Data Received from ECM/PCM A - Value of signal protection calculation incorrect	Missing/invalid data from the engine control module	Using the manufacturer approved diagnostic system, check the engine control module for related DTCs and refer to the relevant DTC index
U0401-87	Invalid Data Received from ECM/PCM A - Missing message	Missing message from the powertrain control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the powertrain control module for related DTCs and refer to the relevant DTC index
U0403-81	Invalid Data Received From Transfer Case Control Module - Invalid serial data received	Invalid data received from transfer case control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index
U0403-86	Invalid Data Received From Transfer Case Control Module - Signal invalid	Invalid data received from transfer case control module	

			NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the transfer case control module for related DTCs and refer to the relevant DTC index
U0415-82	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Alive/sequence counter incorrect / not updated	Missing/invalid data from the anti-lock brake system control module	Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0415-83	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - General checksum failure	Missing/invalid data from the anti-lock brake system control module	Using the manufacturer approved diagnostic system, check the anti-lock brake system control module for related DTCs and refer to the relevant DTC index
U0422-82	Invalid Data Received From Body Control Module - Alive/sequence counter incorrect/not updated	Missing/invalid data from the body control module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the body control module for related DTCs and refer to the relevant DTC index
U0422-83	Invalid Data Received From Body Control Module - Value of signal protection calculation incorrect	Missing/invalid data from the body control module	NOTE: This DTC applies only to hybrid electric vehicles

			Using the manufacturer approved diagnostic system, check the body control module for related DTCs and refer to the relevant DTC index
U0447-82	Invalid Data Received From Gateway "A" - Alive/sequence counter incorrect/not updated	Missing/invalid data from the gateway module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the vehicle supervisory controller/electric power inverter for related DTCs and refer to the relevant DTC index
U0447-83	Invalid Data Received From Gateway "A" - Value of signal protection calculation incorrect	Missing/invalid data from the gateway module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the vehicle supervisory controller/electric power inverter for related DTCs and refer to the relevant DTC index
U0447-87	Invalid Data Received From Gateway "A" - Missing message	Missing/invalid data from the gateway module	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U2101-	Control Module	Transmission	

86	Configuration Incompatible - Signal invalid	control module is not configured correctly	NOTE: This DTC applies only to hybrid electric vehicles Using the manufacturer approved diagnostic system, re-configure the transmission control module with the latest level software. Check and update the car configuration file as necessary. Clear the DTC and re-test
U3000-56	Control Module - Invalid/incomplete configuration	Transmission control module failure	Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new transmission control module
U3001-94	Control Module Improper Shutdown - Unexpected operation	- Transmission control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the manufacturer approved diagnostic system, check datalogger signal - Main ECU Supply Voltage (0xDD02). Refer to the electrical circuit diagrams and test the transmission control module power and ground circuits for open circuit, high resistance - Refer to the relevant section of the workshop manual and test the battery and charging system